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ALGORITHMS IN HISTORY



Most people associate algorithms with computers. This is not unreasonable; computer operating systems use many sophisticated algorithms, and programming is well suited to implementing all sorts of algorithms precisely. But algorithms are more fundamental than the computer architecture we implement them on. As mentioned in Chapter 1, the word *algorithm* dates back about a millennium, and algorithms have been described in ancient records going back much further than that. Even outside of written records, there is abundant evidence for the use of complex algorithms in the ancient world—in, for example, their construction methods.

This chapter presents several algorithms of antique provenance. They show great ingenuity and insight, especially considering that they had to be invented and verified without the aid of computers. We start by discussing Russian peasant multiplication, a method for arithmetic that, despite the